

Pipeline and Vector Processing

SUBJECT:- COA
TOPICS:-Pipeline

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Outline

- ❑ **Parallel processing**
- ❑ **Pipelining**
- ❑ **Arithmetic Pipeline**
- ❑ **Instruction Pipeline**
- ❑ **RISC Pipeline**
- ❑ **Vector Processing**
- ❑ **Array processors**

Parallel Processing

- ❑ To fulfill increasing demands for higher performance it is necessary to process data concurrently to achieve better throughput.
- ❑ Processing data concurrently is known as parallel processing
- ✓ Multiple function unit:- system may have two or more ALU so that they can execute two or more instruction at the same time
- ✓ Multiple processor:- System may have two or more processors operating concurrently.

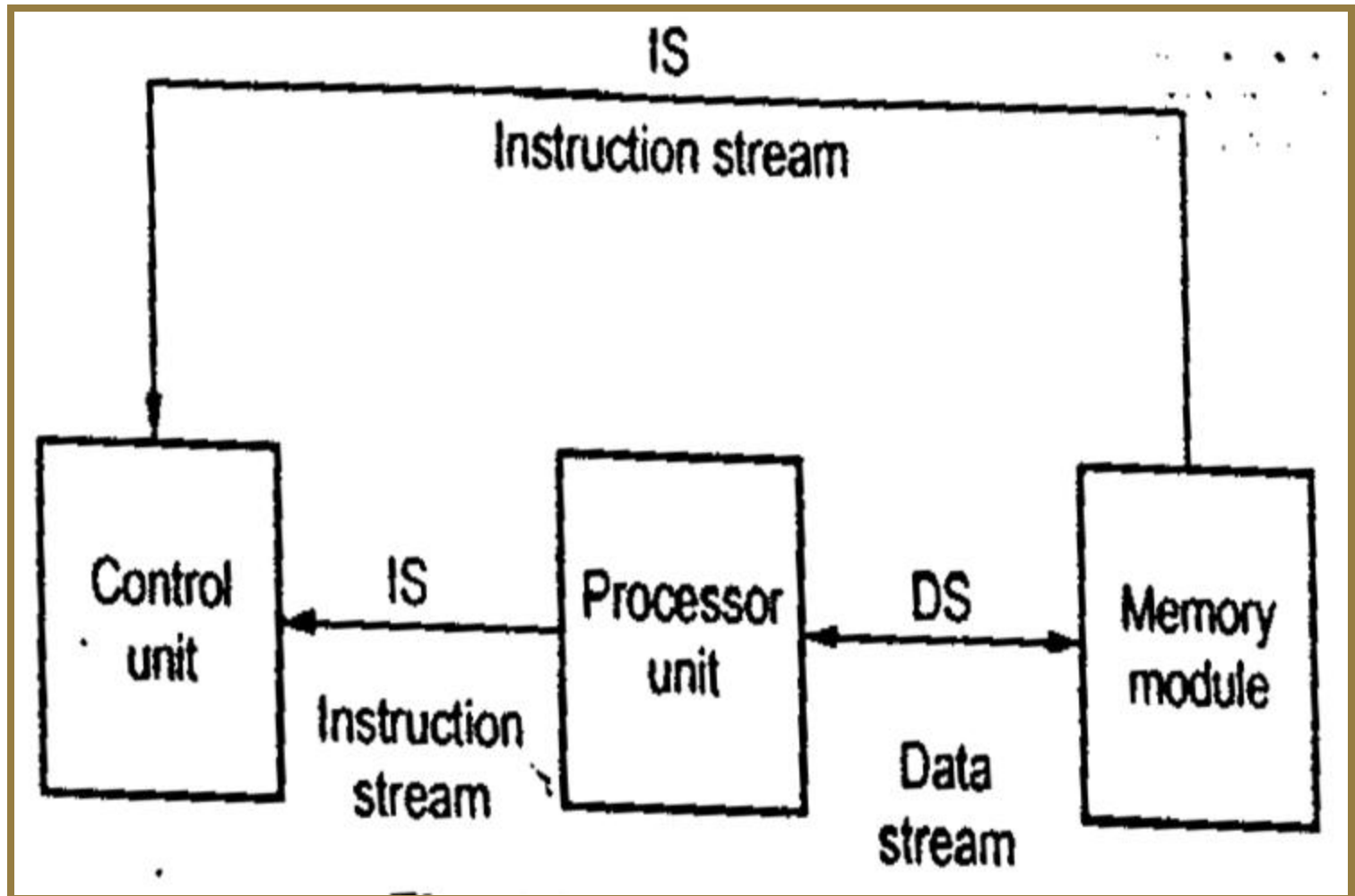
MJ Flynn classification or Parallel Processing classification

- The MJ Flynn classification divide into four group.
 1. Single Instruction stream Single data stream (SISD)
 2. Single Instruction stream Multiple data stream (SIMD)
 3. Multiple Instruction stream Single data stream (MISD)
 4. Multiple Instruction stream Multiple data stream (MIMD)

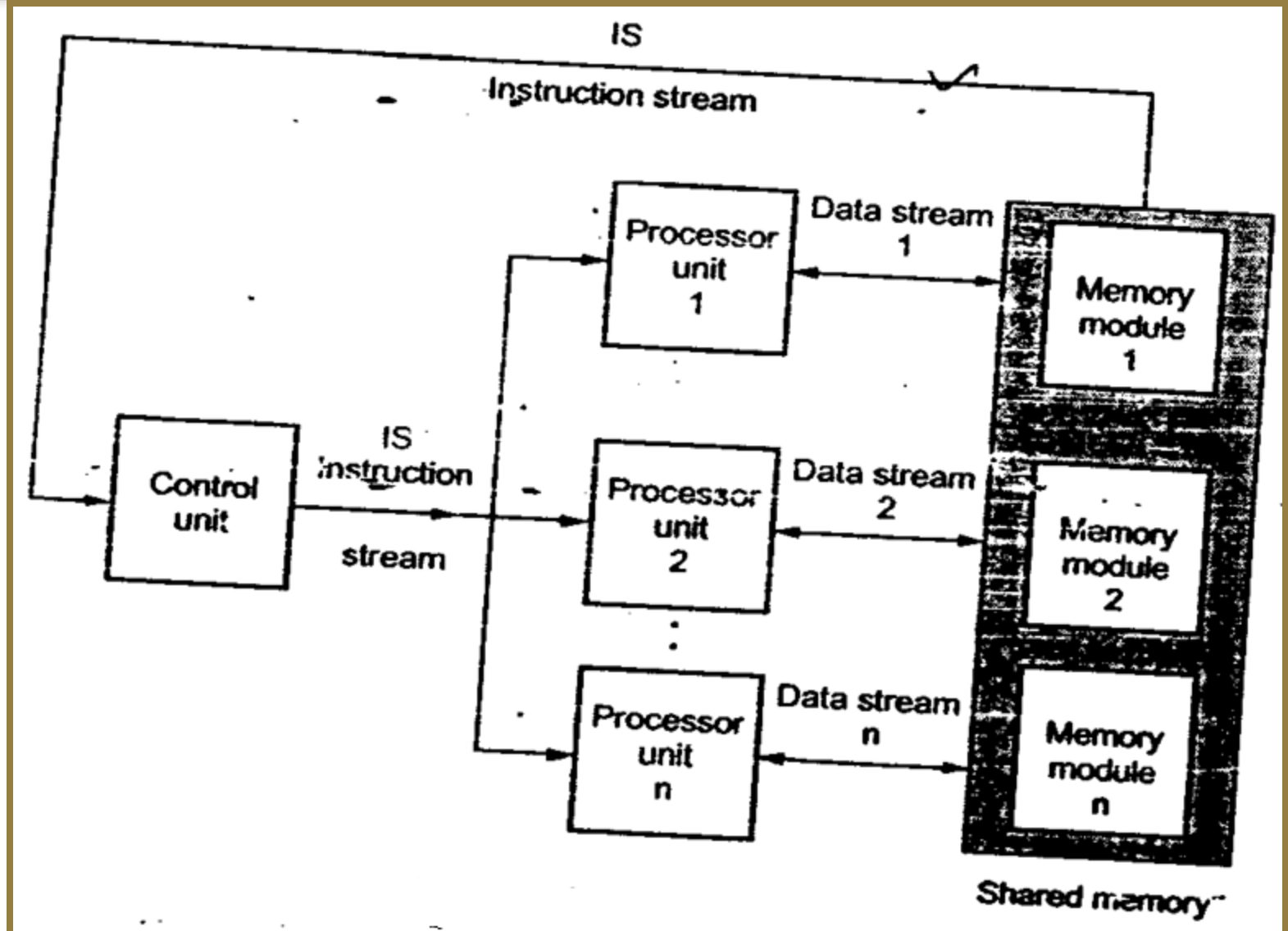
Instruction Stream:- Normal instruction is fetch instruction from memory and execute in processor. This sequence of instruction is an Instruction Stream.

Data stream :- The operation perform on data in the processor or on Data stream.

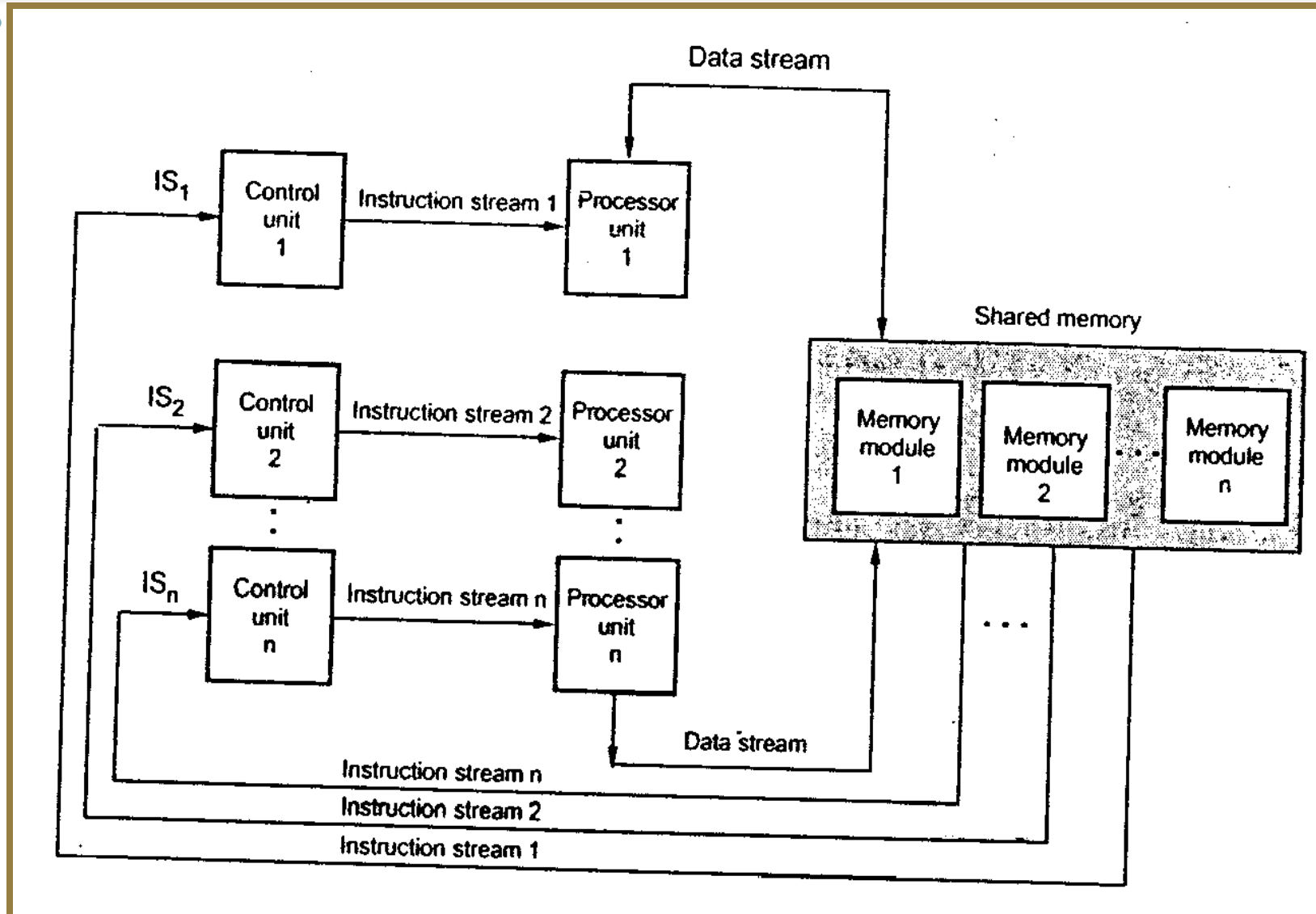
SISD



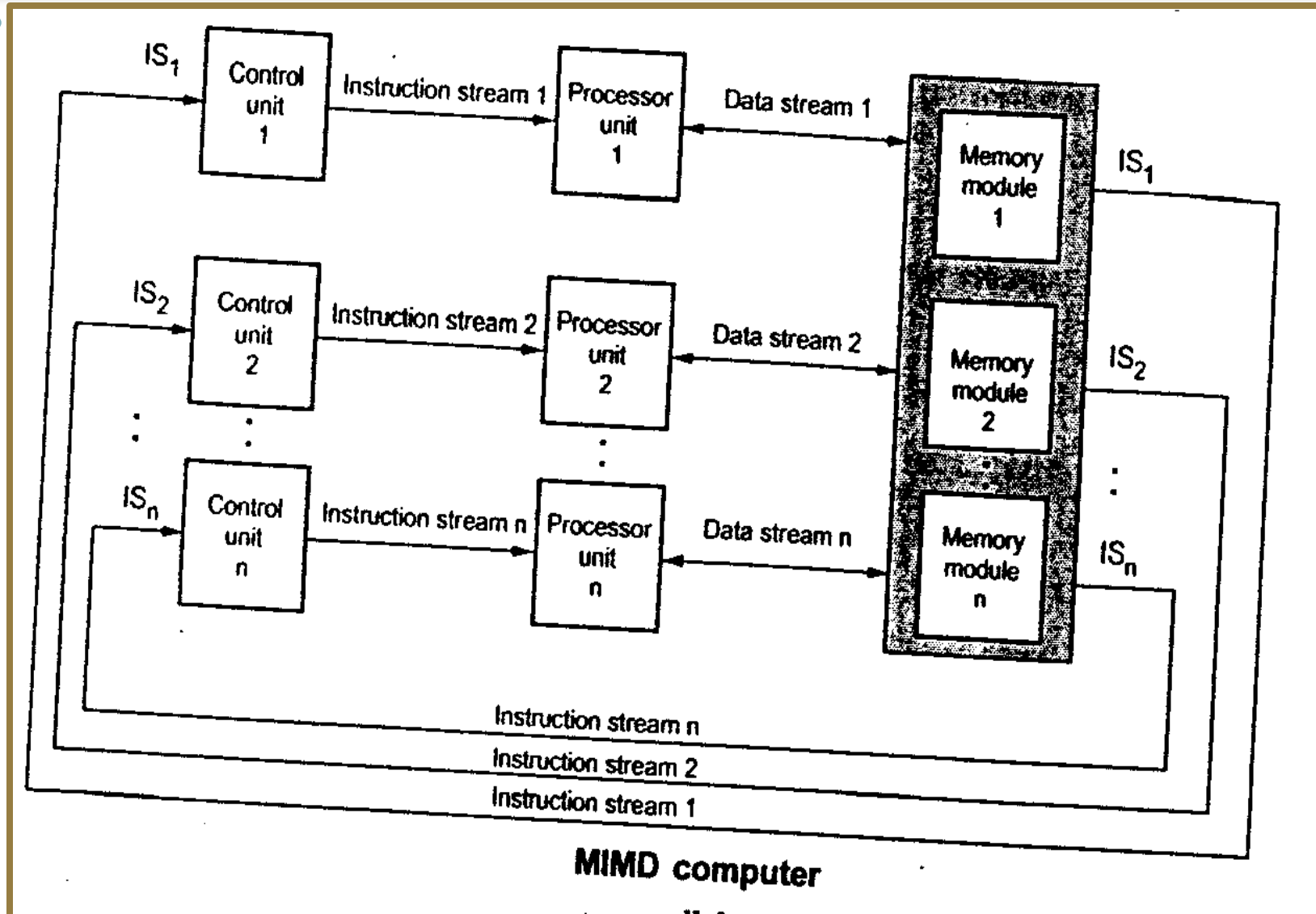
SIMD



MISD



MIMD



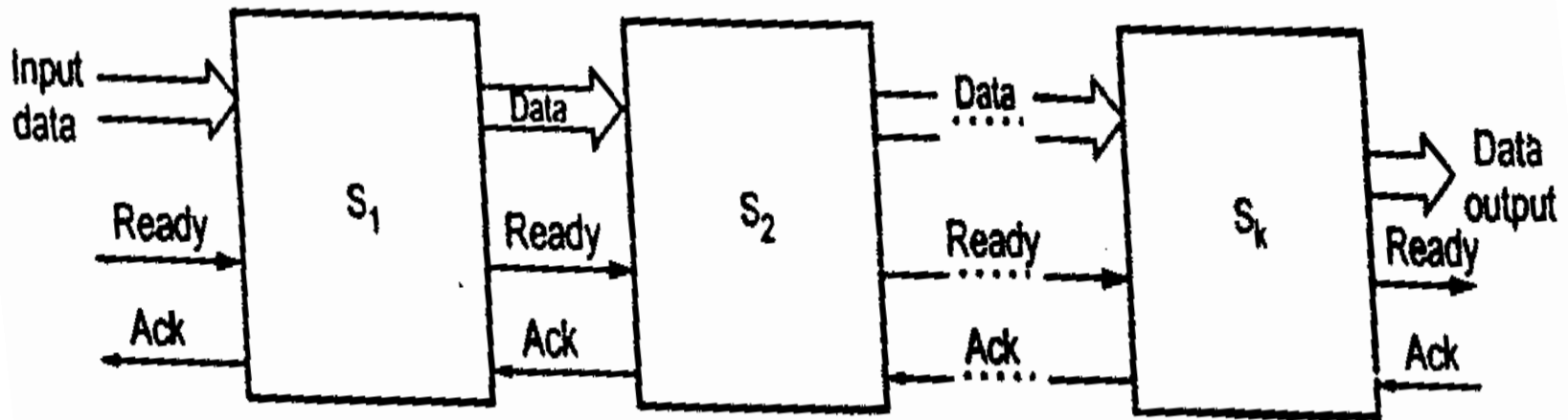
Pipeline processing

- **Pipeline Processing** :- perform overlapped computations or the phases of computer instruction cycle to exploit temporal parallelism.

- **Pipeline** :- The term pipeline refers to the temporal overlapping of processing
 - ✓ Asynchronous pipeline model
 - ✓ Synchronous pipeline model

Asynchronous pipeline model

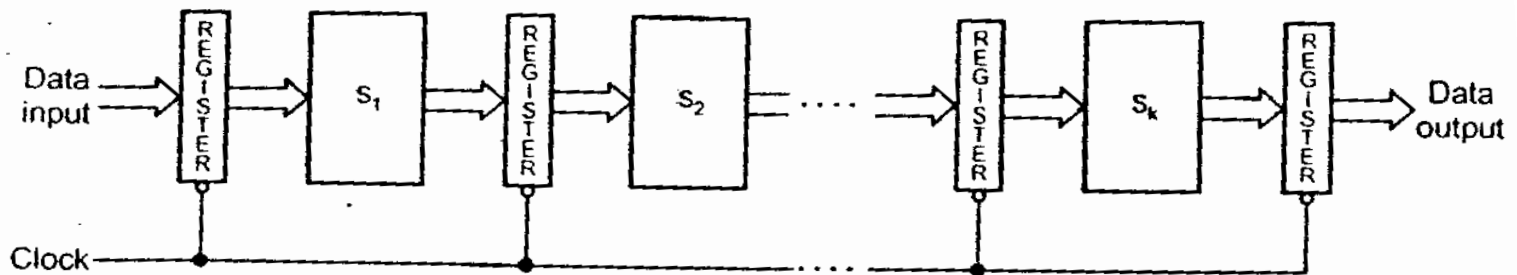
- In asynchronous pipeline model, data flow along the pipeline stages is controlled by Handshaking protocol.



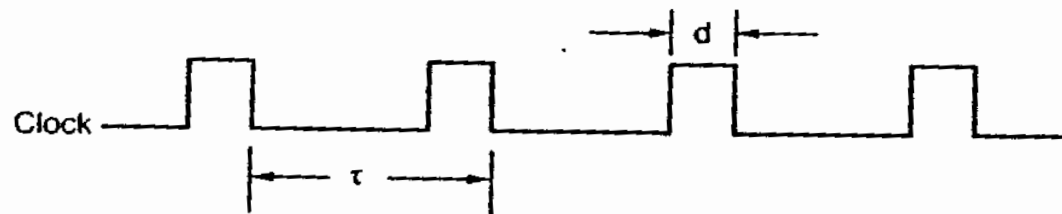
Asynchronous pipeline model

synchronous pipeline model

- In synchronous pipeline model, clock high speed register are used to interface between the stages.



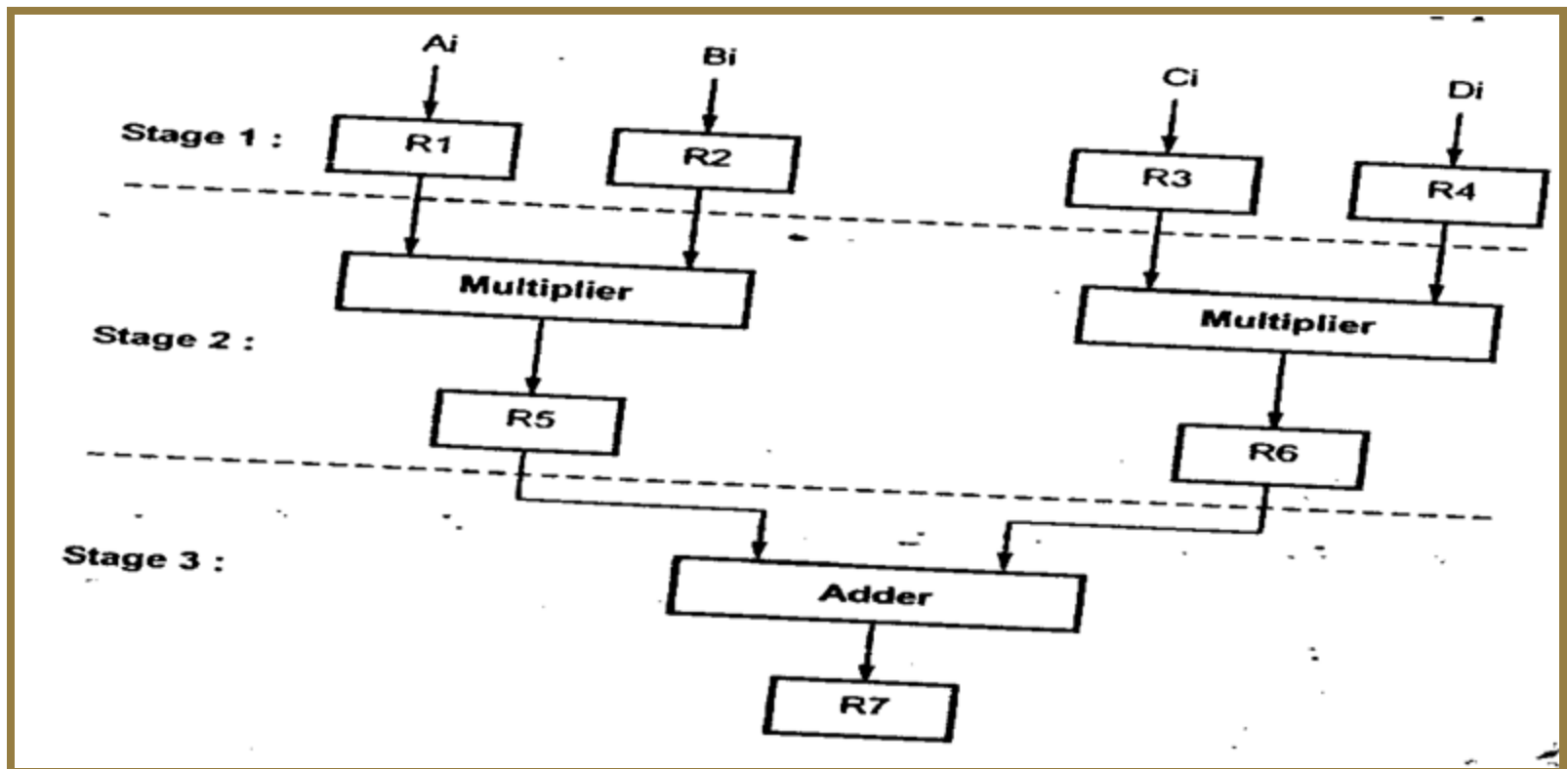
(a) Synchronous pipeline model



(b) Clock signal

Example

- Perform the arithmetic operation $(A_i * B_i) + (C_i * D_i)$ with the stream of number. Specify a pipeline configuration to carry out the task. List the content of all register in the pipeline for $i=1$ through 6.



Cont'd..

Clock Pulse Number (i)	Stage 1				Stage 2		Stage 3
	R1	R2	R3	R4	R5	R6	R7
1	A1	B1	C1	D1	–	–	–
2	A2	B2	C2	D2	$A1 * B1$	$C1 * D1$	–
3	A3	B3	C3	D3	$A2 * B2$	$C2 * D2$	$A1 * B1 + C1 * D1$
4	A4	B4	C4	D4	$A3 * B3$	$C3 * D3$	$A2 * B2 + C2 * D2$
5	A5	B5	C5	D5	$A4 * B4$	$C4 * D4$	$A3 * B3 + C3 * D3$
6	A6	B6	C6	D6	$A5 * B5$	$C5 * D5$	$A4 * B4 + C4 * D4$
7	–	–	–	–	$A6 * B6$	$C6 * D6$	$A5 * B5 + C5 * D5$
8	–	–	–	–	–	–	$A6 * B6 + C6 * D6$

Contents of all registers in the pipeline for $i = 1$ through 6

Example

- Draw a space-time diagram for a six-segment pipeline showing the time for T_i (task) it takes to process eight task.

ANSWER

segment	1	2	3	4	5	6	7	8	9	10	11	12	13
1	T1	T2	T3	T4	T5	T6	T7	T8					
2		T1	T2	T3	T4	T5	T6	T7	T8				
3			T1	T2	T3	T4	T5	T6	T7	T8			
4				T1	T2	T3	T4	T5	T6	T7	T8		
5					T1	T2	T3	T4	T5	T6	T7	T8	
6						T1	T2	T3	T4	T5	T6	T7	T8

Arithmetic pipeline

- ❑ A digital computer perform fixed-point or integer arithmetic or floating point arithmetic.
- ❑ let us consider floating point adder or subtract with two normalize floating input binary A & B.

$$A = M_A * 2^e A$$

$$B = M_B * 2^e B$$

To perform addition or subtraction we have to perform following operation

- ✓ Exponent comparison
- ✓ Chose large exponent
- ✓ Add/sub mantissas
- ✓ Normalize the result

Example

- let us consider that we have to add number A & B.

$$A = M_A * 2^e A = 0.10110 * 2^3$$

$$B = M_B * 2^e B = 0.11010 * 2^4$$

Step-1:- Exponent Comparison

$$(3-4) = 1 \text{ difference}$$

Step-2 :- Chose large exponent

$$B = 0.11010 * 2^4$$

$$\text{Now, } A = 0.10110 * 2^3$$

$$= 0.010110 * 2^4$$

Cont'd..

Step:-3 Add mantissas

$$\begin{aligned} A + B &= 0.010110 * 2^4 + 0.11010 * 2^4 \\ &= 2^4 (0.010110 + 0.11010) \\ &= 2^4 (1.001010) \end{aligned}$$

Step:- 4 Normalize the result.

$$\begin{aligned} \text{Result} &= 2^4 * 1.001010 \\ &= 0.1001010 * 2^5 \end{aligned}$$

Another example

$$x = 0.9504 * 10^3$$

$$y = 0.8200 * 10^2$$

$$\text{Result} = 0.103240 * 10^4$$

Example

- ✓ The time delay of the four segment are as follows. $t_1 = 50\text{ns}$, $t_2 = 80\text{ns}$, $t_3 = 90\text{ns}$, $t_4 = 80\text{ns}$. The inter stage delay time is $d = 10\text{ns}$. So major total time period for cycle and speed up.

- ✓ So, time cycle t = maximum of $t_i + d$
$$= 90\text{ns} + 10$$
$$= 100\text{ns}$$

$$\begin{aligned}\text{Total time cycle } T &= t_1 + t_2 + t_3 + t_4 + d \\ &= 50 + 80 + 90 + 80 + 10 \\ &= 310\text{ns}\end{aligned}$$

$$\begin{aligned}\text{Speed up} &= T/t = 310/100 \\ &= 3.1\end{aligned}$$

Example..

- ✓ The time delay of the four segment are as follows. $t_1 = 60\text{ns}$, $t_2 = 70\text{ns}$, $t_3 = 100\text{ns}$, $t_4 = 80\text{ns}$. The inter stage delay time is $d = 10\text{ns}$. So major total time period for cycle and speed up.

Example..

- ✓ The time delay of the four segment are as follows. $t_1 = 40\text{ns}$, $t_2 = 25\text{ns}$, $t_3 = 85\text{ns}$, $t_4 = 55\text{ns}$. The inter stage delay time is $d = 5\text{ns}$. So major total time period for cycle and speed up. In addition find out the how long would it take to add 75 pairs of numbers in the pipeline?

$$\begin{aligned}\text{Clock cycle} &= 85 + 5 \\ &= 90 \text{ ns}\end{aligned}$$

$$\text{No of cycle require of 75 pairs} = k + n - 1$$

$$k\text{- no of stages} = 4 + 75 - 1$$

$$n\text{- no of pair} = 78$$

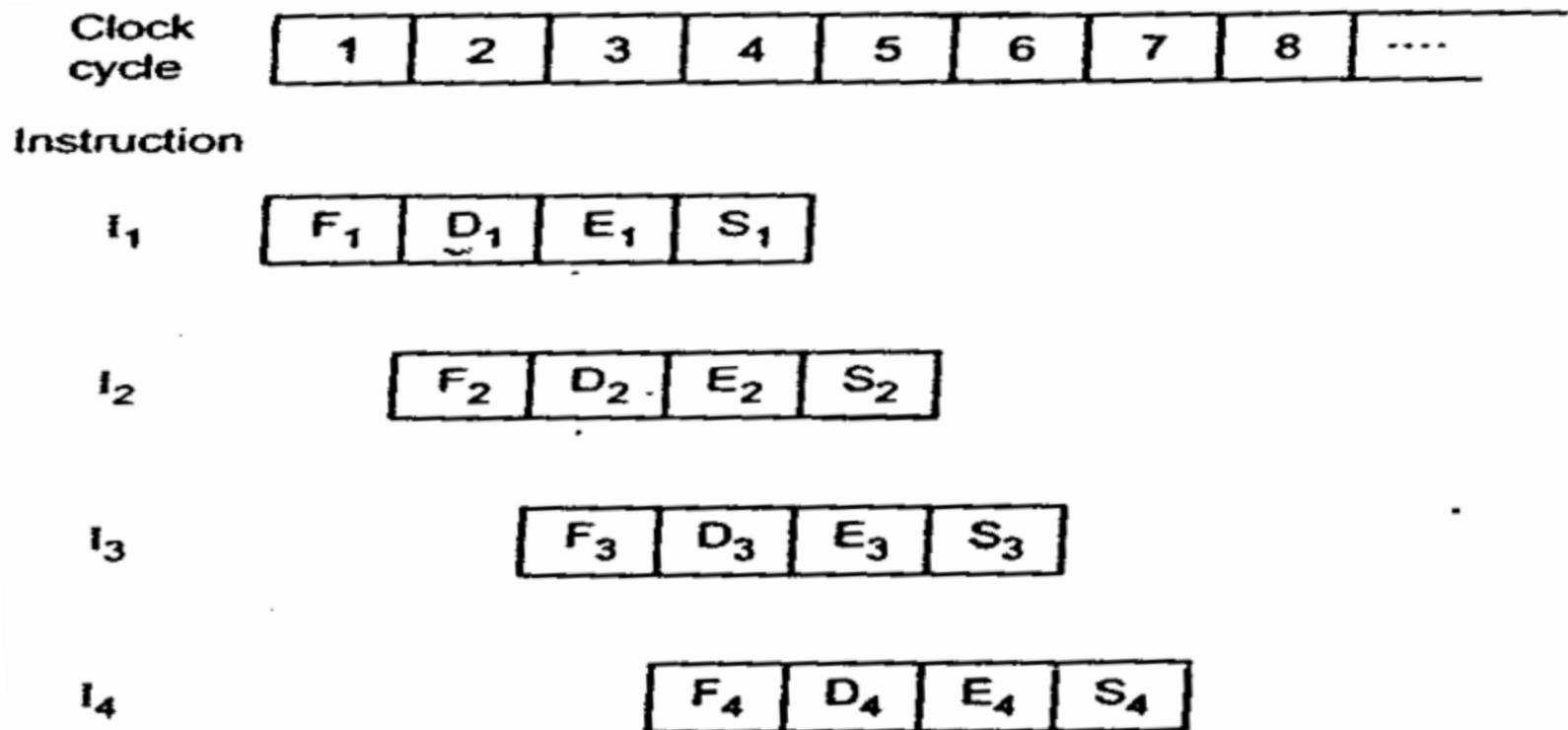
$$\text{Time require to execute 75 pairs} = 78 * 90 = 7020\text{ns}$$

Instruction pipeline..

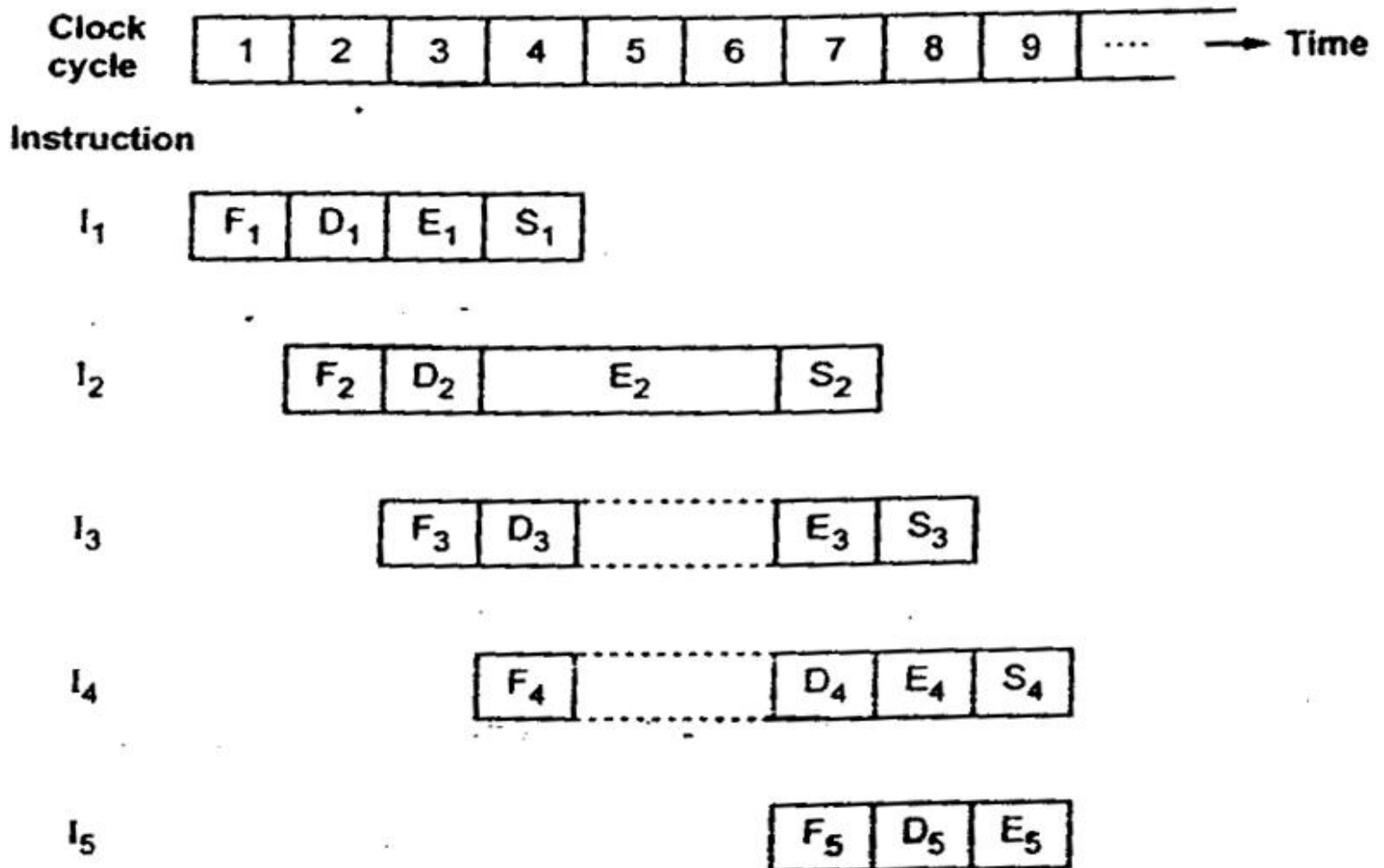
- ✓ **Instruction cycle perform following operation**
 - ✓ **Fetch (F)**
 - ✓ **Decode (D)**
 - ✓ **Execute (E)**
 - ✓ **Store (S)**
-
- ✓ **Instruction cycle involves various cycles. These sub cycles are for several instruction are performed simultaneously to reduce overall processing time. This process is referred to as instruction pipelining.**

Four Stage Instruction pipeline

- ✓ Instruction processing is divided into four stage hence it is known as four stage instruction pipeline



Pipeline conflict



Pipeline conflict

- ✓ The normal four stage instruction pipeline operation is finished in 7 clock cycle.
- ✓ Stall:- The instruction pipeline shown in fig is said to have been stalled for two clock cycle.
- ✓ Hazard or Conflict:- Any reason that cause the pipeline to stall is called a Hazard or conflict

Types of Pipeline conflict

- ✓ Resources conflicts :- This conflict are happen due to insufficient resources. Like not sufficient memory to store data.
- ✓ Data or data dependency conflict:-These result when instruction in the pipeline depend on the result of previous instruction which is not completed.
- ✓ Branch difficulties:- They arise while pipeline branch and other instruction that change the content of program counter. Like $(A+B)*C$

Techniques to avoid data conflict

- ✓ Hardware Interlock / Bubble:- The easiest way to handle data conflict is to stall the pipeline. Such as printer, telephone.

- Printer:- 1 person can print his document while other have to wait.

- Telephone:- while 2 person talking 3rd person cant talk.

- Bubble:- delay is called bubble.

- ✓ Operand Forwarding:- The second simple hardware technique. In this technique ALU result is transfer in ALU input through the path instead of ALU output because H/W checks if the result is used for next instruction.

Explain Following terms

° Pipeline scheduling:- Rearrange the instruction

Branch difficulties/conflict:- These is cause due to overlapped instruction in pipeline. Branch conflict can reduce the performance. These is occurs due to branch or jump which change the flow of instruction .

Branch penalty:- It is depend upon the depth of pipeline. More deep pipeline, there is more branch penalty. Branch penalty is also depend upon the set of instruction of computer (CISC)

Delayed branch:- delay slot is filled from the target branch instruction.

Conditional branching

- ✓ Multiple streams:- In this approach, it is used two stream to store the fetch instruction
- ✓ Prefetch branch target:- In this approach ,when a conditional branch is recognized, the target of the branch is pre fetched.
- ✓ Loop buffer:- A loop buffer is a small very high speed memory. It is used to store recently pre fetched instruction in sequence.
- ✓ Branch prediction:- prediction techniques can be used to check whether a branch will be valid or not valid.

Vector Processing

- ✓ The need to increase computational power is a never ending requirement. In scientific and research areas, the computation involved are quite extensive and hence high power computers are a must.
- ✓ Areas like structural engineering, Petroleum exploration, aerodynamics, nuclear research, VLSI design, etc.
- ✓ Examples:- Radar, weather forecasting, medical diagnosis, image processing, etc.

Characteristic of vector processing

- ✓ A vector is order set of element $A \rightarrow B$
- ✓ Faster process in one clock cycle.
- ✓ Vector instruction are classified into four basic type
 - f1: $V \rightarrow V$
 - f2: $V \rightarrow S$
 - f3: $V * V \rightarrow V$
 - f4: $V * S \rightarrow V$
- ✓ One process repeat many times and divide in sub process

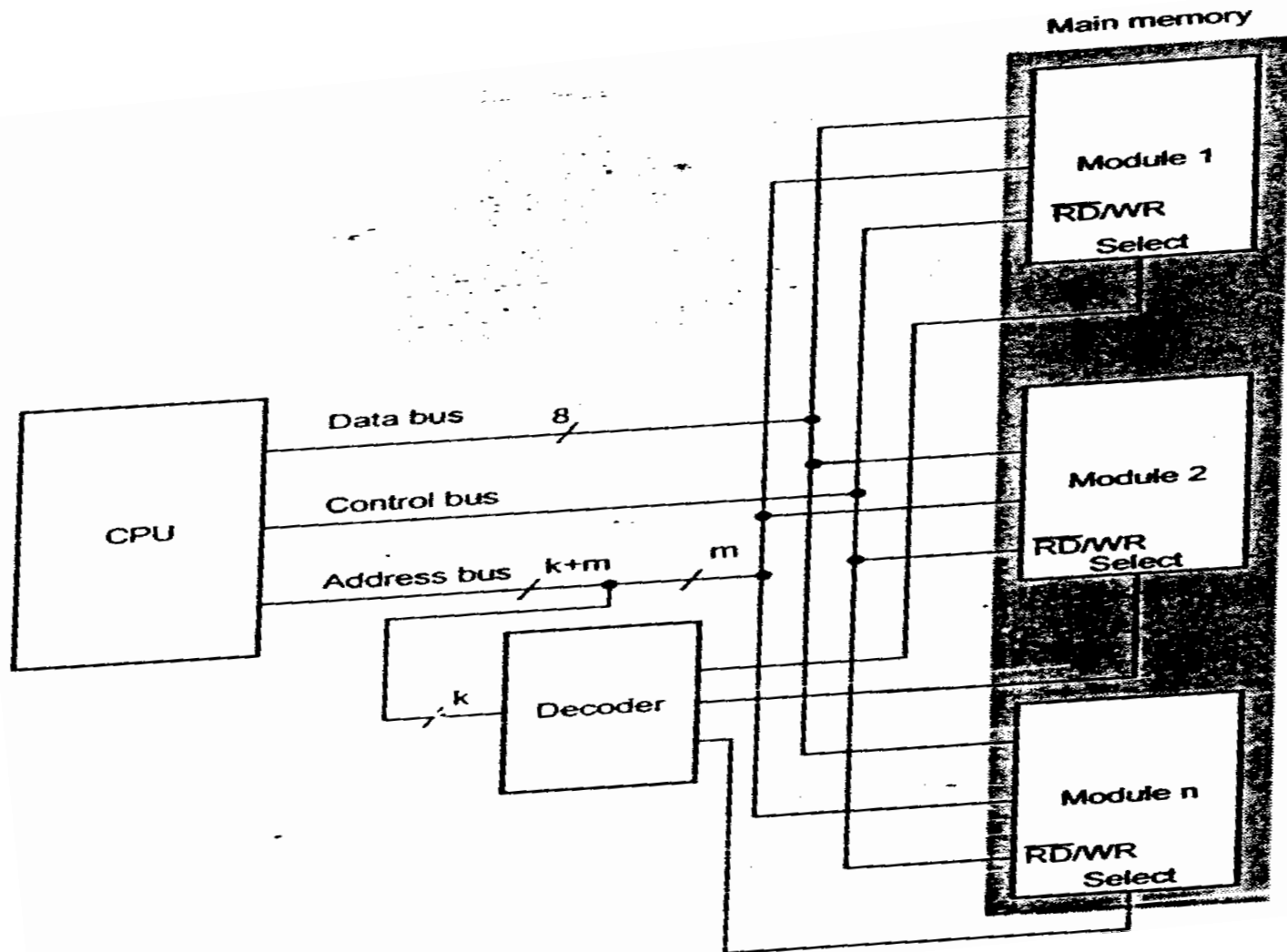
Vector instruction field

Op code	Base address source 1	Base address source 2	Base address destination	Address Increment	Address offset	Vector length
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Memory Interleaving

- ✓ Pipeline and vector processor often require simultaneous access to memory from two or more sources.
- ✓ An instruction pipeline may require the fetching of opcode and operand at the same time from two different memory segment.
- ✓ In such system, the memory access time is the bottleneck (small space) and one way to reduce it is to use cache memory.
- ✓ Alternative technique to reduce memory access time is memory interleaving.
- ✓ In this technique, the main memory is divided into number of memory modules and the addresses are arranged in sequence manner.

Memory interleaving



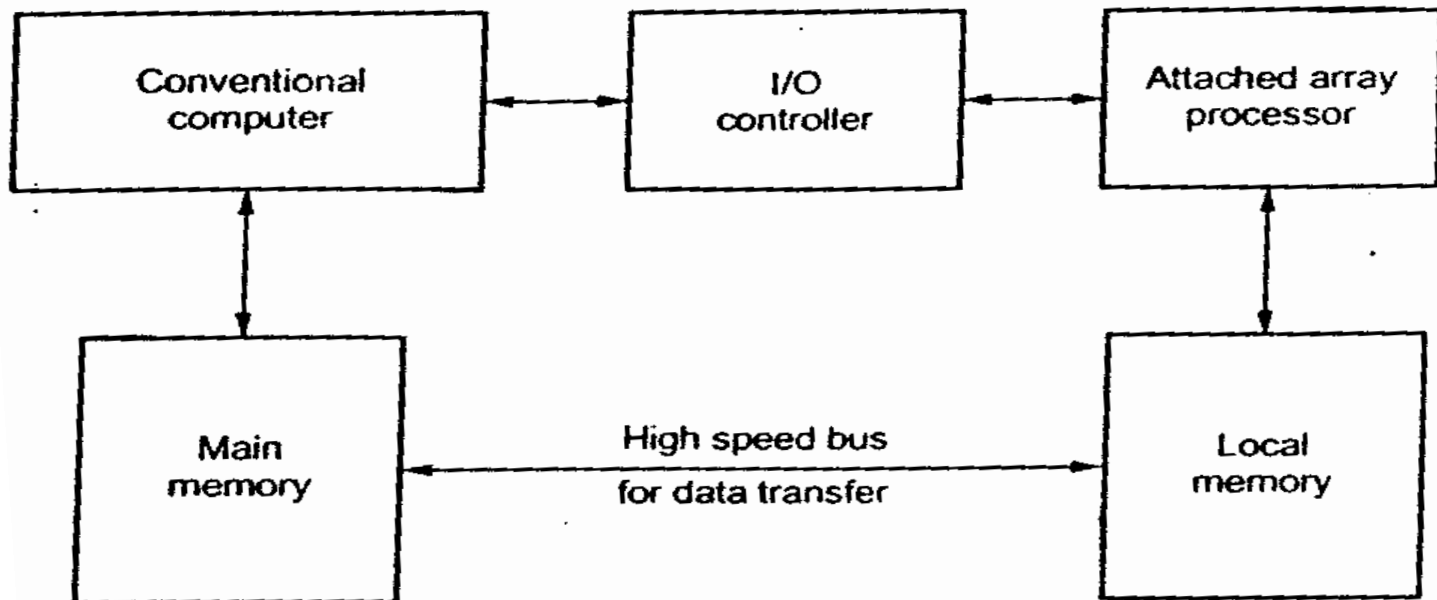
Supercomputers

- ✓ Capacity of performing large-scale vector or matrix computation.
- ✓ It is used for structural engg, VLSI, AI, nuclear research, etc.
- ✓ The performance of high speed computer is evaluated in terms of FLOPS(number of floating point operation per second).
- ✓ Ability to perform Megaflop(million flop) and Gigaflop(billion flop)
- ✓ CRAY-1 and CRAY-2 are examples of supercomputers.

Array processor.

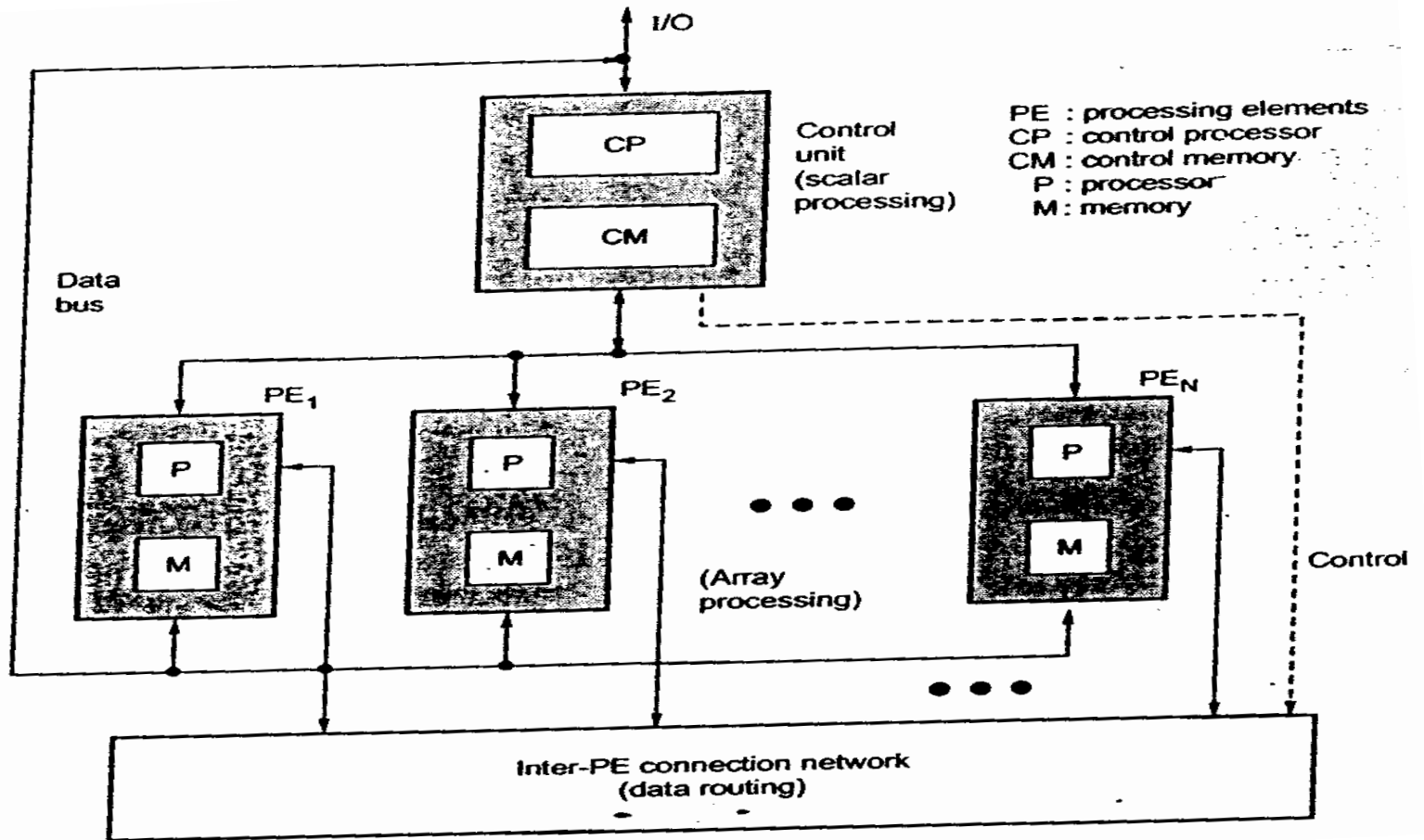
- ✓ An array processor is synchronous parallel computer with multiple arithmetic logic units.
- ✓ These arithmetic logic units are also known as processing element(PE).
- ✓ Array processors means perform computation on large array of data.
- ✓ There are two types of array processors.
 - Attached array processor
 - SIMD array processor

Attached array processor



Attached array processor with conventional computer

SIMD array processor





Thank You !

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